
Observability, Memory, and the Dynamics of Cooperation in Heterogeneous Multi-Agent Reinforcement Learning

Mariana Meireles*

Department of Theoretical Evolutionary Biology
Max Planck Institute
Plön, Germany
meireles@evolbio.mpg.de

Wolfram Barfuss

Center for Development Research
University of Bonn
Bonn, Germany
wbarfuss@uni-bonn.de

Abstract

We ask how three design choices—what an agent *observes*, how much history it *remembers*, and which *learning rule* it uses—shape whether reciprocal cooperation emerges in multi-agent reinforcement learning. Using collective reinforcement-learning dynamics (CRLD), the deterministic limit of temporal-difference learning, we read the learning process as a *flow* in policy space and study a heterogeneous Iterated Prisoner’s Dilemma in which one agent observes the full history and the other is selectively blind along controlled dimensions. Three findings emerge. First, *self-observability is the load-bearing perceptual channel*: an agent that cannot see its own last action cannot implement a reciprocal rule and the flow collapses to mutual defection, whereas losing sight of the *partner*—or seeing it only conditionally—is far less damaging. Second, *the cooperative basin is created by memory*: absent at memory one, a cooperative attractor appears at memory two and enlarges as agents condition on more history, a purely deterministic counterpart to the intrinsic-fluctuation route to cooperation. Third, *the learning rule reshapes the attractor*: policy-gradient learning is bistable between full cooperation and full defection, whereas value-based learning settles on a robust interior compromise that avoids total defection—the same game yields different cooperative outcomes under the two rules. Throughout, the dominant failure mode of observational heterogeneity is *shared* defection rather than one agent dominating another. Reading the dynamics as a flow makes each of these mechanisms—and the basins behind them—explicit.

1 Introduction

Cooperation underlies much of human flourishing Hare and Woods [2020], Bar-Yam [2000], yet the complexity of modern societies routinely produces social dilemmas in which individual incentives and collective welfare diverge—from climate policy Kersting [2017] and pandemic response Glennerster et al. [2024] to political stalemate Frieden and Lake [2015], Dafoe et al. [2020] and economic distrust Acemoglu and Robinson [2012]. Such problems are increasingly modeled with Markov Decision Processes and reinforcement learning (RL), which capture how agents—governments, firms, consumers—adapt strategies from partial observations of an evolving environment Glennerster et al. [2024], Charpentier et al. [2021].

A defining feature of real-world actors is that they do *not* perceive their shared environment identically: a firm and a regulator see different slices of the same market, and treaty negotiators hold asymmetric

*<https://marimeireles.com>

information about each other—and about their own commitments. We therefore ask a focused question: *how does heterogeneity in observability shape whether learning agents come to cooperate?* We isolate observability from other heterogeneities (learning rate, risk tolerance) so its effect can be measured cleanly, in the general-sum Iterated Prisoner’s Dilemma Barfuss [2022].

Our approach is to treat the learning process itself as a dynamical system. In the deterministic CRLD limit, the learning rule is a vector field (a *flow*) on the space of joint policies; cooperation corresponds to a basin of attraction, and we can ask directly which design choices create or destroy that basin. This lets us move past a single cooperation statistic to the *geometry* that produces it, and to sweep observability, memory, and learning rule on the same footing. A preliminary report by the present authors over-read these simulations as showing that knowledge-asymmetric agents cooperate readily; the controlled, flow-based analysis here gives a sharper and more mechanistic account.

2 Methods

2.1 Learning dynamics

We model the two agents, $i \in \{1, 2\}$ choosing $a^i \in \{c, d\}$, with *collective reinforcement learning dynamics* (CRLD) Barfuss [2023, 2022], Barfuss et al. [2019]: the deterministic, infinite-batch limit of temporal-difference learning, which replaces stochastic value updates with an exact map in policy space. We use this machinery as given—it is prior work—and recall only what we need. Each agent’s policy $x^i(a \mid o)$ (over actions a given an *observation* o) evolves by a replicator–mutator step along its strategy-averaged temporal-difference error δ^i :

$$x_{t+1}^i(a \mid o) = \frac{x_t^i(a \mid o) \exp(\alpha_i \beta_i \delta_t^i(o, a))}{\sum_b x_t^i(b \mid o) \exp(\alpha_i \beta_i \delta_t^i(o, b))}, \quad (1)$$

with learning rate α_i , choice intensity β_i , and discount γ_i . The error δ^i is the standard expected, on-policy TD error (immediate reward plus discounted successor value minus baseline; the infinite-batch limit of Expected SARSA Sutton and Barto [2018]), whose closed form we take unchanged from Barfuss et al. [2019], Barfuss and Mann [2022]. These parameters can differ per agent; we hold them equal ($\alpha = 0.1$, $\gamma = 0.9$, $\beta = 1$) so that observability is the sole heterogeneity.

Heterogeneous observations. Our modification is to make the agents heterogeneous *observers*. The true state $s \in \mathcal{S}$ is the joint two-round history, but agent i acts on an observation o^i drawn from a *per-agent* observation model $\Omega^i(o^i \mid s) \in \{0, 1\}$ that may alias several true states onto one observation. A full observer uses the identity model ($o^i \equiv s$); a partial observer’s Ω^i masks chosen coordinates of the most recent round, leaving the agent with the uniform posterior over the states consistent with what it saw,

$$b^i(s \mid o^i) = \frac{\Omega^i(o^i \mid s)}{\sum_{s'} \Omega^i(o^i \mid s')}, \quad (2)$$

and every strategy-average entering δ^i is taken under b^i Barfuss and Mann [2022]. *Heterogeneity is precisely the choice of different Ω^i across agents*: here Agent 1 keeps the full model while Agent 2’s Ω^2 realizes one of the masking regimes of Section 2.3. Concretely, we extend pyCRLD to carry a separate observation tensor per agent rather than a single shared one.

The dynamics as a flow. Equation (1) defines a deterministic map on the joint policy—equivalently, a vector field (a *flow*) on strategy space. From any joint policy the learning rule points in a definite direction, the fixed points where the flow vanishes are the learned equilibria, and the *cooperation rate* we report is the measure of the basin of attraction of the cooperative fixed point under random initial policies. Visualizing this flow (Figures 1–2) exposes *why* a scenario does or does not cooperate, complementing the reward statistics with the geometry behind them.

2.2 Environment

Our focal environment is the Iterated Prisoner’s Dilemma (IPD), whose one-shot payoffs (Table 1) are $R = 1$ (mutual cooperation), temptation $T = 1.2$, sucker $S = -0.5$, and punishment $P = 0$, satisfying $T > R > P > S$ and $2R > T + S$. The Prisoner’s Dilemma is, however, only one of the four canonical symmetric two-action social dilemmas: reordering the same four payoffs yields *Harmony* (cooperation is dominant), the *Stag Hunt* (a bistable coordination game), and *Snowdrift/Chicken* (anti-coordination). We run our full analysis—observability, memory, and learning rule—across all four games, so that we can separate effects driven by the Prisoner’s Dilemma’s specific incentive structure from those that hold across dilemmas; the exact payoffs of the other three games are given in the Appendix (Table 2). Results in the main text are reported for the IPD unless stated otherwise.

We embed each stage game into a *memory-two* history (an agent’s state is the previous two rounds’ outcomes)—the smallest history rich enough to host the reciprocal strategies (win-stay–lose-shift, tit-for-tat) that direct reciprocity relies on Schmid et al. [2021], Hilbe et al. [2017], yet low-dimensional enough for exhaustive analysis.

These matrix games are also the incentive cores of richer, spatially and temporally extended multi-agent environments. Two that we study come from the JaxMARL suite Rutherford et al. [2024]: the *coin game*, in which two agents move on a grid and pick up own- versus other-coloured coins—a spatial Prisoner’s Dilemma—and the in-the-matrix *arena* (STORM), in which N agents forage coloured resources and, on meeting, play a matrix game set by their inventories—a temporally extended, N -player social dilemma. The coin game’s incentive core is the Prisoner’s Dilemma analyzed here, and the arena generalizes it to $N > 2$, letting us ask whether the observability and learning-rule effects survive beyond the dyad.

Table 1: Payoff matrix for the Iterated Prisoner’s Dilemma (row agent, column agent).

	c	d
c	(1, 1)	(-0.5, 1.2)
d	(1.2, -0.5)	(0, 0)

2.3 Observational heterogeneity

Agent 1 is a *full observer* (perceives both agents’ last two actions without aliasing). Agent 2 is a *partial observer*: along controlled dimensions, parts of the most recent round are masked, aliasing several true states to one observation (its belief is spread uniformly over them) while the prior round stays fully observed. We study four restrictions of Agent 2; exact state-to-observation maps are in the Appendix (Tables 3–6).

- **Self-aware:** Agent 2 remembers its *own* most recent action but not Agent 1’s (it cannot see what the other did last round).
- **Non-self-aware:** Agent 2 remembers Agent 1’s most recent action but not its *own* (it cannot see what it itself did last round).
- **Cooperation-tracking:** Agent 2 perceives Agent 1’s last action only when Agent 1 *cooperated*; Agent 1’s defections are masked.
- **Defection-tracking:** Agent 2 perceives Agent 1’s last action only when Agent 1 *defected*; Agent 1’s cooperations are masked.

We probe each setting two ways: by the *flow geometry* in policy space (Figure 1), and by integrating the deterministic dynamics from $N = 1,200$ random softmax initial policies to a fixed point (tolerance 10^{-5} , $\leq 4,000$ steps). The *cooperation rate* is the fraction of runs reaching mutual cooperation (both cooperation probabilities > 0.9)—the measure of the cooperative basin under random initialization.

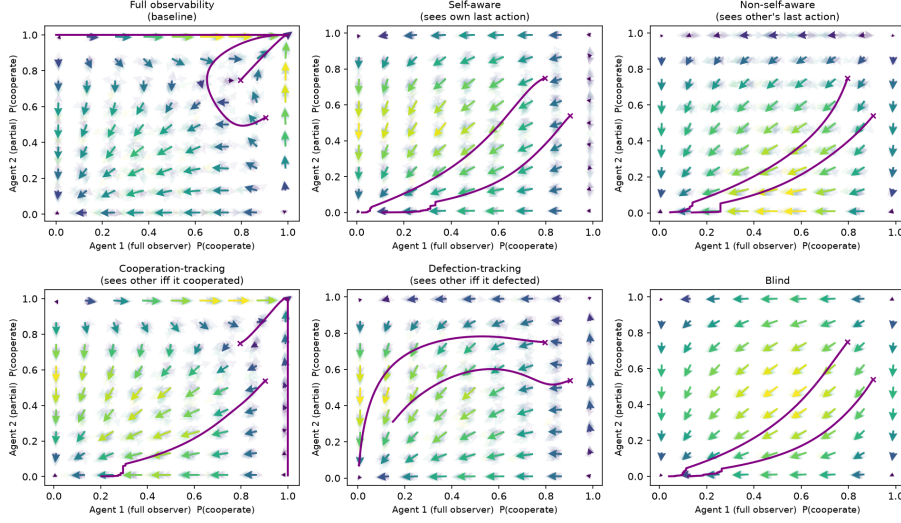


Figure 1: Learning flow of the heterogeneous memory-two IPD (actor-critic CRLD), one panel per observability condition of the partial observer (Agent 2; Agent 1 is always a full observer). Arrows are the deterministic learning update at each joint policy; purple curves are two trajectories; axes are each agent’s cooperation probability. Mutual cooperation is the corner $(1, 1)$, defection $(0, 0)$. A basin draining to cooperation exists under full observability, cooperation-tracking, and—weakly—self-awareness, but the flow collapses toward defection when the partial observer cannot see its *own* last action (non-self-aware) or is blind.

3 Results

Self-observability is the load-bearing perceptual channel. Reading the heterogeneous IPD as a flow (Figure 1) shows cooperation living in a basin that observability controls. Under full observability a cooperative fixed point at $(1, 1)$ is reachable, but it shares the space with a large defection basin—about 14% of random initializations cooperate—so even symmetric, complete information does not make cooperation typical. Degrading the partial observer shrinks the cooperative basin, and *which* coordinate is hidden matters far more than how much is hidden. When Agent 2 cannot see its *own* last action (non-self-aware), the flow drains uniformly to defection: an agent that cannot identify its own move cannot “stay” or “shift” relative to it, so the substrate for direct reciprocity is gone. Losing sight of the *partner* instead (self-aware) is far milder—a cooperative basin survives—and merely *conditioning* which of the partner’s moves are visible (cooperation- vs. defection-tracking) is close to full observability. It is self-observability, not partner-observability, that reciprocity is built on. Across initializations the two agents converge to nearly the same policy, so the failure mode of heterogeneity is *shared* defection rather than one agent exploiting the other.

Memory creates and enlarges the cooperative basin. That cooperative basin is not intrinsic to the game—it is built by *memory*. Re-running the full-observability dynamics at memory one, two, and four (Figure 2) shows that at memory one the flow does not reach mutual cooperation from our initializations; a cooperative fixed point at $(1, 1)$ becomes reachable only at memory two, and its basin enlarges at memory four. Longer histories let agents represent richer reciprocal strategies (memory- k tit-for-tat / win-stay-lose-shift) that turn cooperation into a deterministic attractor. This is the deterministic counterpart of the finite-batch *intrinsic fluctuations* known to promote cooperation in the memory-one IPD Barfuss and Meylahn [2023]: memory and sampling noise enlarge the same cooperative basin by different routes.

The learning rule co-determines the attractor. Whether cooperation is reachable at all also depends on the learning rule (Appendix, Figure 3). Policy-gradient (actor-critic) learning is *bistable*: depending on initialization it converges either to mutual cooperation $(1, 1)$ or to mutual defection $(0, 0)$, with most random starts in the defection basin. Value-based (SARSA) learning instead

Full observability: the cooperative basin emerges and grows with memory (IPD, actor-critic)

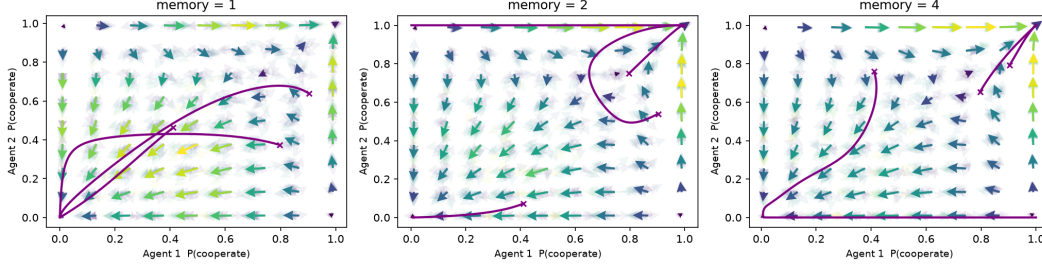


Figure 2: Full-observability IPD learning flow at memory one, two, and four (actor-critic). A cooperative fixed point at $(1, 1)$ is not reached at memory one but becomes an attractor from memory two onward, and its basin enlarges with memory—a deterministic analogue of the intrinsic-fluctuation mechanism Barfuss and Meylahn [2023].

converges, from almost any start, to a single *interior* partial-cooperation fixed point—it never collapses to total defection, but never reaches full cooperation either. The same game, payoffs, and observability thus produce qualitatively different cooperative outcomes under the two rules, so the choice of algorithm is part of the cooperation question, not an implementation detail. (In Snowdrift/Chicken the gap is starker still—the two rules select different equilibria outright, a point we return to in Section 5.)

The effects survive beyond the dyad and the matrix game. The observability and learning-rule effects are not artifacts of the two-player matrix game. In the N -player arena, analyzed in the same deterministic limit, self-observability remains the load-bearing channel (Appendix, Figure 4), and cooperation need not fall as agents are added—in this pairwise-averaged dilemma it can even rise with N . The findings also transfer out of the deterministic limit: the same heterogeneous IPD learned by *sampled* deep RL reproduces the algorithm ordering—value-based IQL climbs to near-full cooperation while policy-gradient IPPO collapses to defection (Appendix, Figure 5)—while the spatial, temporally extended coin game and arena Rutherford et al. [2024] make cooperation harder overall, with a pronounced collapse at large N (Appendix, Figure 6).

4 Discussion

Reading multi-agent learning as a flow turns “does cooperation emerge?” into the sharper question of *what shapes the cooperative basin*, and our three results answer it with three distinct knobs. *Observability* sets whether reciprocity is expressible at all: self-observation is the load-bearing channel, because a reciprocal rule is defined relative to one’s own last move, while partner-observation is comparatively dispensable. *Memory* sets whether a cooperative attractor exists: it is absent at memory one and built up as agents condition on more history. *The learning rule* sets the shape of the attractor: policy-gradient is bistable between full cooperation and full defection, whereas value-based learning settles on a robust interior compromise.

A unifying reading is that cooperation here is gated by how much reciprocal structure the learners can both *represent* and *stabilize*. Memory and finite-batch intrinsic fluctuations Barfuss and Meylahn [2023] expand the same basin by different routes—more representable history versus noise that escapes the defection trap—which suggests they are partly interchangeable; whether fluctuations can rescue the non-self-aware case that memory alone cannot is a natural test. Equally, the dominant failure mode throughout is *shared* defection rather than exploitation: perceptual asymmetry costs both agents the cooperative surplus instead of letting the better-informed one dominate.

5 Future work and limitations

The result we find most provocative—and least understood—is *algorithm-driven equilibrium selection*. In Snowdrift/Chicken, value-based learning reliably settles on the symmetric, mutually cooperative

mixed equilibrium, whereas policy-gradient is drawn instead to the asymmetric anti-coordination outcomes: one learning rule discovers a “good” cooperative solution that the other simply does not find. We have no mechanistic account of *why* a learning rule prefers one equilibrium over another, nor of which rules induce cooperative selection in which games. A predictive theory of this would matter well beyond our setting—it would tell a designer which learner to deploy to obtain a desired social outcome—and we think it is the most promising thread to pull next.

Beyond it, natural steps are to (i) sweep β and γ to map where observational heterogeneity becomes cooperation-promoting; (ii) reintroduce finite-batch fluctuations Barfuss and Meylahn [2023] to test whether they rescue the non-self-aware case that memory alone cannot; and (iii) push the deep-RL coin game and arena Rutherford et al. [2024] further, and on to language-model agents. The present conclusions are correspondingly bounded: the matrix games capture a narrow slice of strategic behavior; CRLD assumes deterministic, infinite-batch update rules rather than human idiosyncrasy; outcomes depend on the chosen parameters and the regimes we report; and cooperation rates are statements about the measure of a basin of attraction under random initial policies, not about any single trajectory.

6 Acknowledgments

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A Appendix / supplemental material

A.1 Software and reproducibility

All experiments use an extension of the open-source pyCRLD library Barfuss [2023] that adds heterogeneous per-agent observation tensors. The deterministic dynamics, observation-tensor construction, and figure-generation scripts used for every result and figure in this paper are released with the camera-ready version.

Relation to an earlier preliminary report. An earlier exploratory writeup by the authors reported that knowledge-asymmetric agents cooperate readily and that better-informed agents earn more. Those conclusions do not survive a controlled analysis: the optimistic cooperation reading came partly from a run with an out-of-range discount factor ($\gamma = 10$, which makes the $(1 - \gamma)$ prefactor negative and the value solve ill-posed) and from per-scenario reporting drift, and the apparent reward advantage of the better-informed agent vanishes—indeed reverses—once a valid $\gamma = 0.9$ is used. All results here use one consistent, flow-based pipeline at $\gamma = 0.9$.

A.2 Parameters

For all experiments we use learning rate $\alpha = 0.1$, discount factor $\gamma = 0.9$, and choice intensity $\beta = 1$, identical across both agents. Memory length is two ($h = (2, 2, 2)$ in the history-embedding convention) except in the memory sweep (Figure 2), which varies it. Each scenario uses $N = 1,200$ random softmax initial policies; trajectories run up to 4,000 steps with fixed-point tolerance 10^{-5} .

A.3 Game family

The four canonical symmetric two-action social dilemmas studied here share the structure of Table 1 and differ only in the payoff values (Table 2), i.e. in the ordering of R, T, S, P . This ordering determines the equilibrium structure—cooperation-dominant (Harmony), a single defective dominant strategy (Prisoner’s Dilemma), bistable coordination (Stag Hunt), or anti-coordination (Snowdrift/Chicken)—and lets us test which observability, memory, and learning-rule effects are specific to the Prisoner’s Dilemma.

Table 2: Payoffs of the four social dilemmas (mutual cooperation R , temptation T , sucker S , punishment P) and their defining inequality.

Game	R	T	S	P	ordering
Prisoner’s Dilemma	1	1.2	−0.5	0	$T > R > P > S$
Harmony	1	0.5	0.5	0	$R > T, S > P$
Stag Hunt	1	0.5	−0.5	0	$R > T > P > S$
Snowdrift / Chicken	1	1.5	0.5	0	$T > R > S > P$

A.4 Learning rule

A.5 Richer environments: N -player and deep RL

The Results claim that the effects survive beyond the two-player matrix game rests on the following supplementary analyses. Figure 4 repeats the observability flow grid for the N -player arena ($N = 3$, deterministic CRLD): self-awareness remains the load-bearing channel, exactly as in the dyad. Figure 5 leaves the deterministic limit altogether and trains sampled deep-RL agents on the same heterogeneous-observation IPD; the value-based learner (IQL) reaches near-full cooperation while the policy-gradient learners (IPPO, A2C) decay toward defection, reproducing the deterministic algorithm ordering. Figure 6 shows the spatial, temporally extended STORM arena Rutherford et al. [2024], where cooperation is harder to reach and collapses as the number of agents grows to $N = 8$.

Learning rule and cooperation: full-observability IPD flow (memory-2)

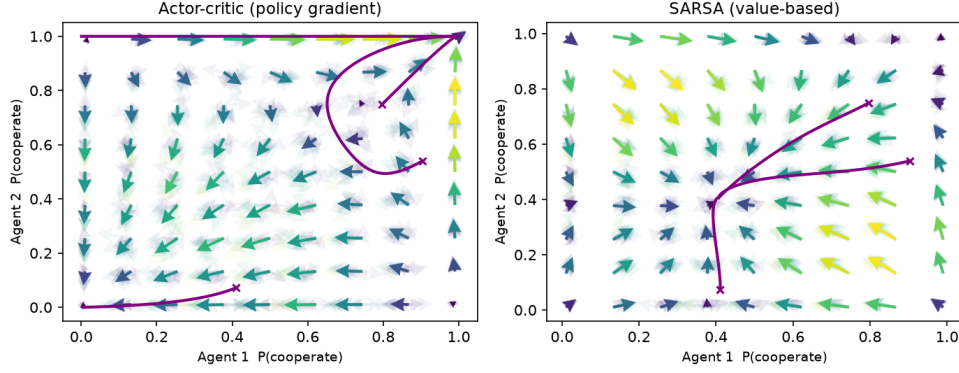


Figure 3: The learning rule reshapes the attractor (full-observability IPD, memory-two). *Left*: policy-gradient (actor-critic) is bistable—initial conditions decide between mutual cooperation (1, 1) and mutual defection (0, 0). *Right*: value-based (SARSA) learning converges from almost any start to a single interior partial-cooperation fixed point, avoiding total defection but never reaching full cooperation.

arena | CRLD actor-critic | $N=3$ | observability flow grid (agents 0,1 projection)

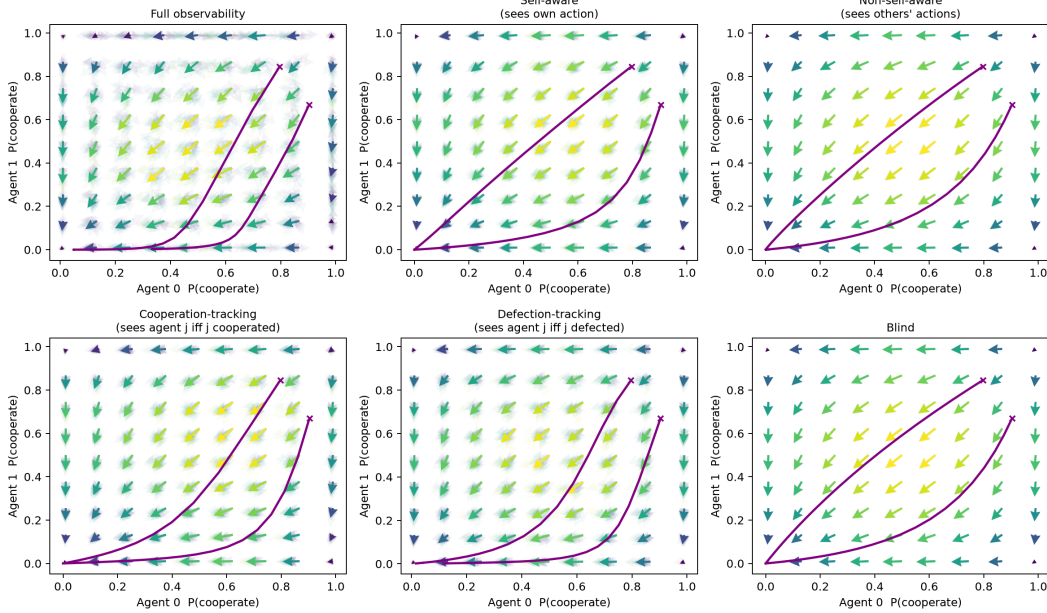


Figure 4: N -player arena ($N = 3$), deterministic CRLD observability flow grid (actor-critic). As in the two-player IPD, self-awareness (top-middle) preserves a cooperative basin while non-self-awareness (top-right) and blindness drain toward defection—the observability finding is not specific to the dyad.

Algorithm comparison on the heterogeneous-observation IPD ($N=2$)

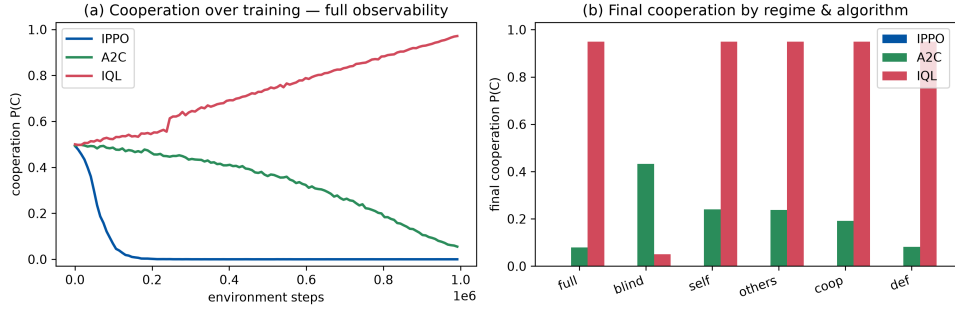


Figure 5: Sampled deep RL on the heterogeneous-observation IPD (away from the deterministic limit). (a) Cooperation over training under full observability: value-based IQL rises toward full cooperation, policy-gradient IPPO collapses to defection, A2C decays slowly. (b) Final cooperation by observability regime and algorithm: the value-based/policy-gradient ordering holds across regimes. This reproduces, under sampling, the algorithm dependence found in the deterministic flow.

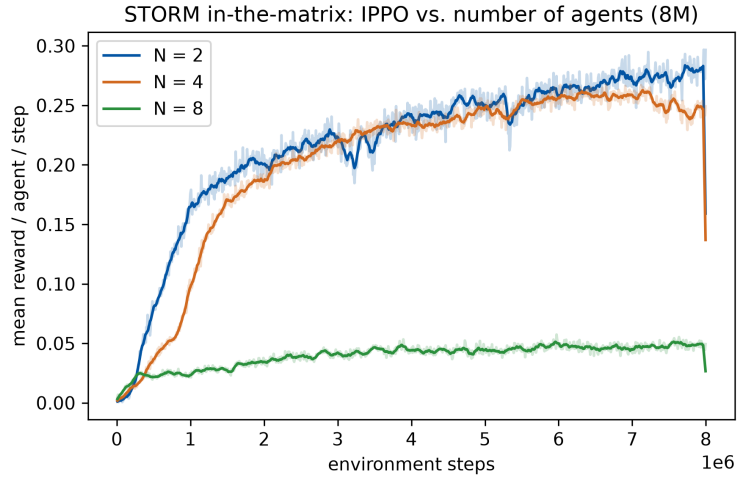


Figure 6: STORM in-the-matrix arena Rutherford et al. [2024] (deep-RL, IPPO): mean reward per agent over training for $N = 2, 4, 8$ agents. Cooperation is harder than in the dyadic matrix game and collapses at $N = 8$ —an effect of the spatial, temporally extended setting that the deterministic dyadic analysis does not capture.

A.6 Observation maps

Tables 3–6 give the state-to-observation maps for Agent 2 in each scenario. Each row is one of the 16 memory-two states; entries show what Agent 2 perceives, with * denoting a masked (unobserved) coordinate that aliases the state with others sharing the same observation. Agent 1 observes every state without aliasing in all scenarios.

Table 3: Self-aware Agent 2: keeps its own most recent action, masks Agent 1’s.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(*, C)
(C, C)	(C, D)	(C, C)	(*, D)
(C, D)	(C, C)	(C, D)	(*, C)
(C, D)	(C, D)	(C, D)	(*, D)
(D, C)	(D, C)	(D, C)	(*, C)
(D, C)	(D, D)	(D, C)	(*, D)
(D, D)	(D, C)	(D, D)	(*, C)
(D, D)	(D, D)	(D, D)	(*, D)

Table 4: Non-self-aware Agent 2: keeps Agent 1’s most recent action, masks its own.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(C, *)
(C, C)	(C, D)	(C, C)	(C, *)
(C, D)	(C, C)	(C, D)	(C, *)
(C, D)	(C, D)	(C, D)	(C, *)
(D, C)	(D, C)	(D, C)	(D, *)
(D, C)	(D, D)	(D, C)	(D, *)
(D, D)	(D, C)	(D, D)	(D, *)
(D, D)	(D, D)	(D, D)	(D, *)

Table 5: Cooperation-tracking Agent 2: sees Agent 1’s last action only when Agent 1 cooperated.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(C, C)
(C, C)	(C, D)	(C, C)	(C, D)
(C, C)	(D, C)	(C, C)	(*, C)
(C, C)	(D, D)	(C, C)	(*, D)
(D, C)	(C, C)	(D, C)	(C, C)
(D, C)	(C, D)	(D, C)	(C, D)
(D, C)	(D, C)	(D, C)	(*, C)
(D, C)	(D, D)	(D, C)	(*, D)

Table 6: Defection-tracking Agent 2: sees Agent 1’s last action only when Agent 1 defected.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(*, C)
(C, C)	(C, D)	(C, C)	(*, D)
(C, C)	(D, C)	(C, C)	(D, C)
(C, C)	(D, D)	(C, C)	(D, D)
(D, C)	(C, C)	(D, C)	(*, C)
(D, C)	(C, D)	(D, C)	(*, D)
(D, C)	(D, C)	(D, C)	(D, C)
(D, C)	(D, D)	(D, C)	(D, D)